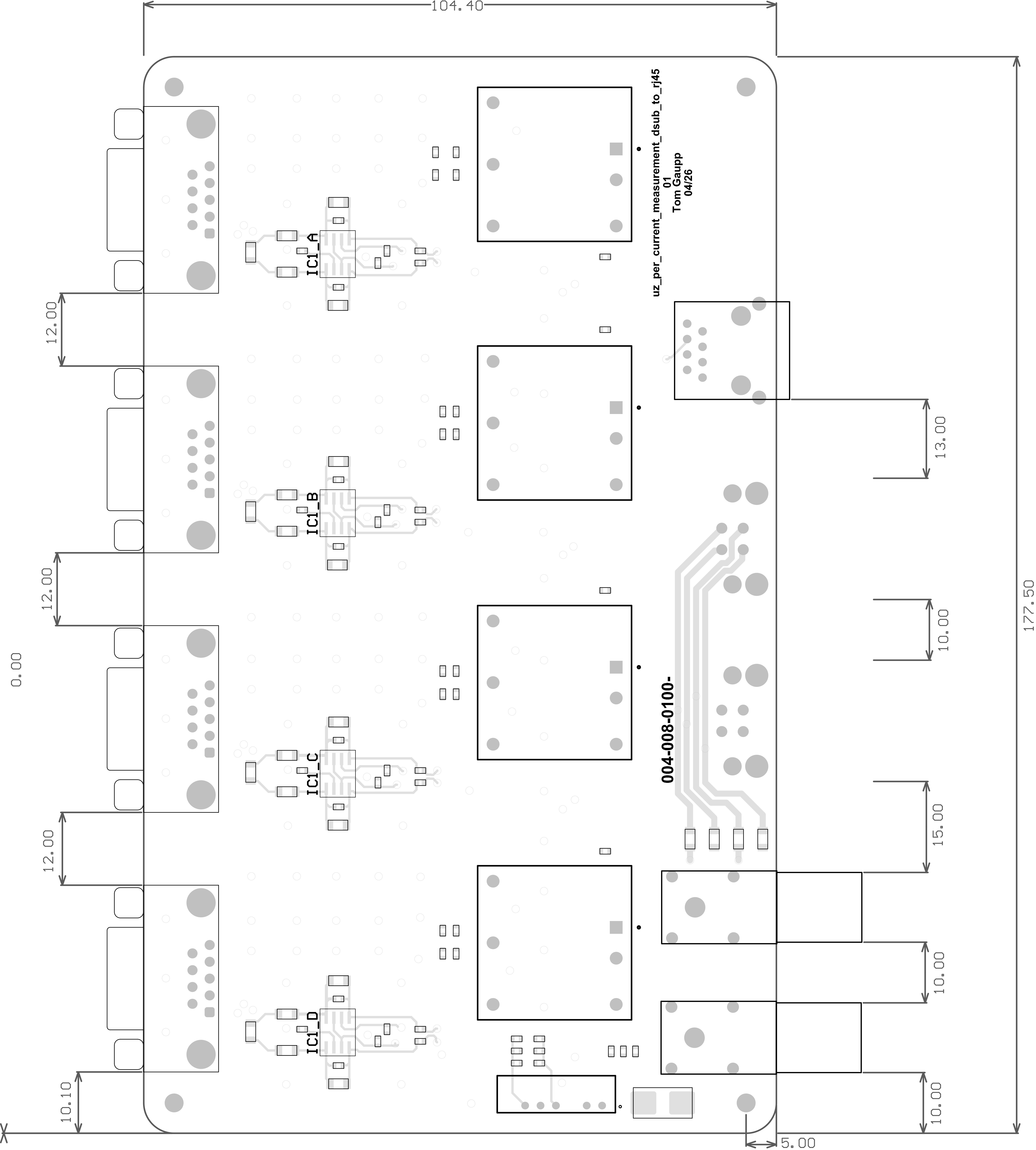
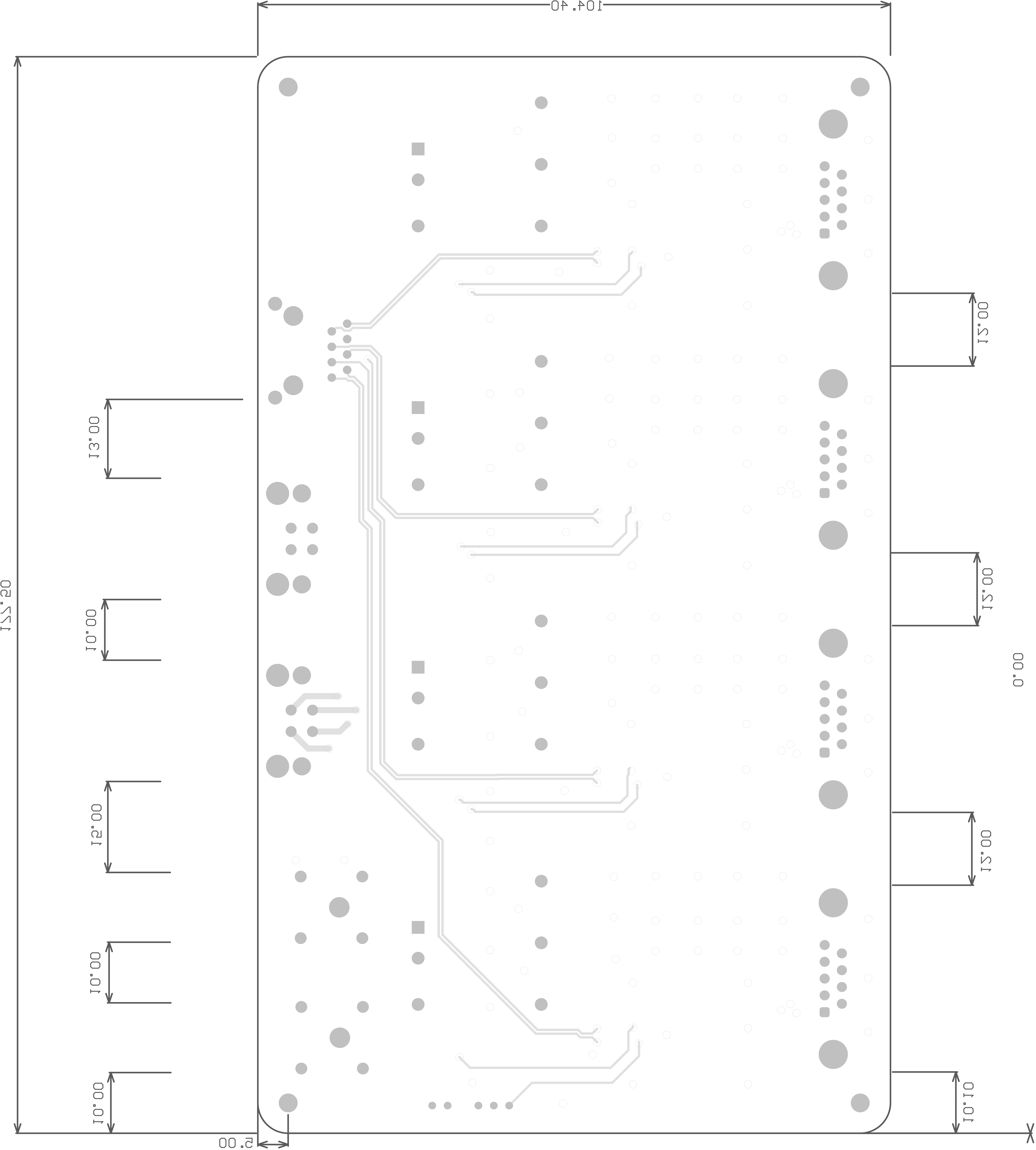
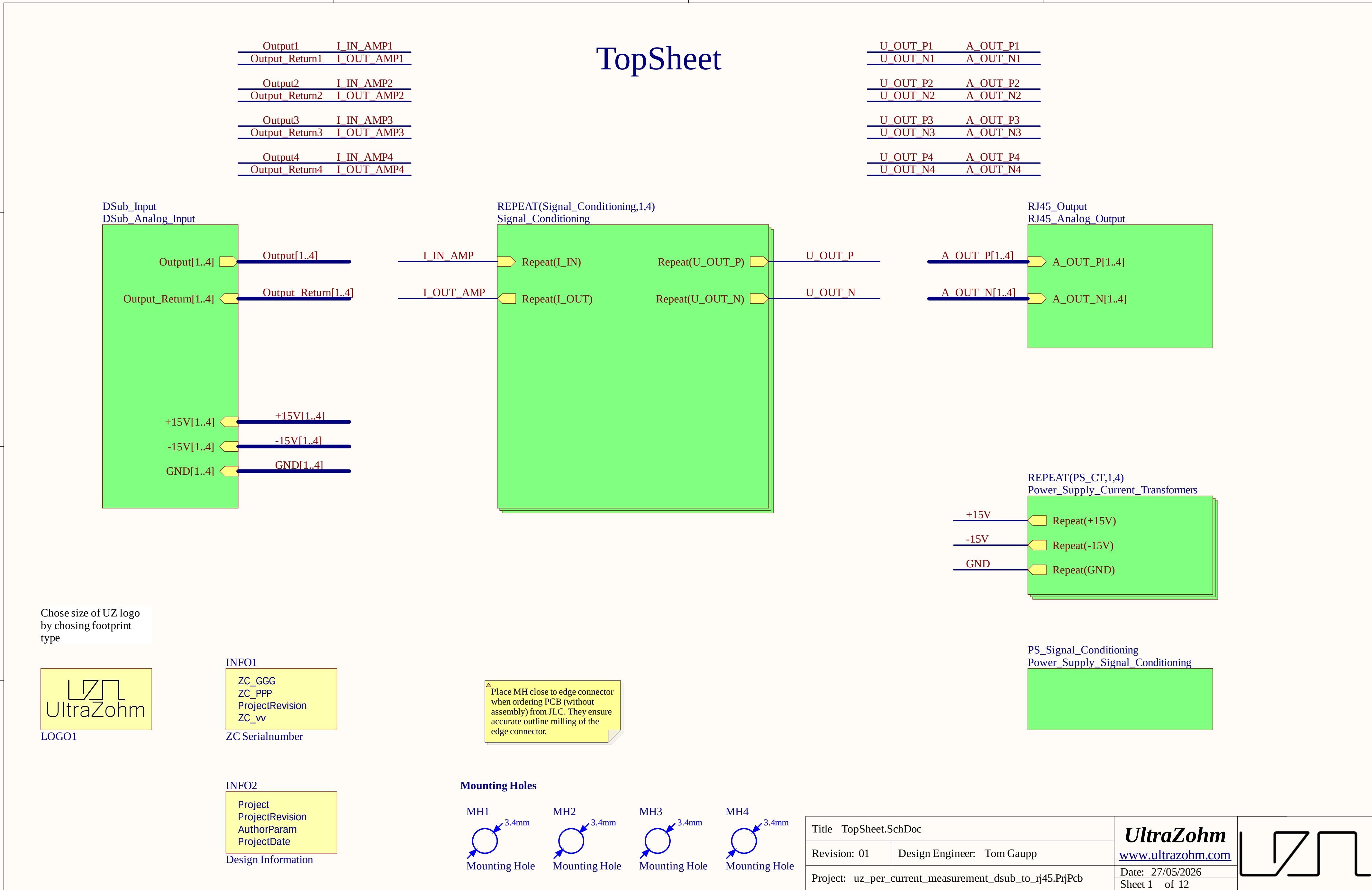
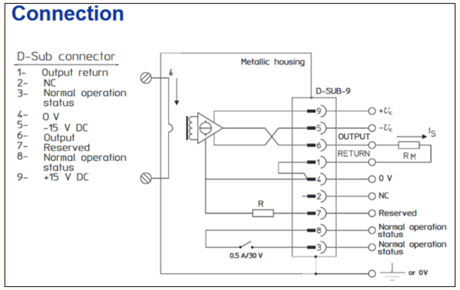

104.40
uz_per_current_measurement_dsub_to_rj45
01
Tom Gaupp
04/26
IC1_A
IC1_B
IC1_C
IC1_D
12.00
12.00
12.00
0.00
10.10
13.00
10.00
15.00
10.00
10.00
177.50
5.00
004-008-0100-


104.40
122.50
13.00
10.00
15.00
10.00
10.00
5.00
12.00
12.00
12.00
10.10
0.00


TopSheet
Output1 I_IN_AMP1
Output_Return1 I_OUT_AMP1
Output2 I_IN_AMP2
Output_Return2 I_OUT_AMP2
Output3 I_IN_AMP3
Output_Return3 I_OUT_AMP3
Output4 I_IN_AMP4
Output_Return4 I_OUT_AMP4
U_OUT_P1 A_OUT_P1
U_OUT_N1 A_OUT_N1
U_OUT_P2 A_OUT_P2
U_OUT_N2 A_OUT_N2
U_OUT_P3 A_OUT_P3
U_OUT_N3 A_OUT_N3
U_OUT_P4 A_OUT_P4
U_OUT_N4 A_OUT_N4
DSub_Input
DSub_Analog_Input
Output[1..4]
Output_Return[1..4]
+15V[1..4]
-15V[1..4]
GND[1..4]
REPEAT(Signal_Conditioning,1,4)
Signal_Conditioning
I_IN_AMP
I_OUT_AMP
Repeat(I_IN)
Repeat(I_OUT)
Repeat(U_OUT_P)
Repeat(U_OUT_N)
U_OUT_P
U_OUT_N
RJ45_Output
RJ45_Analog_Output
A_OUT_P[1..4]
A_OUT_N[1..4]
REPEAT(PS_CT,1,4)
Power_Supply_Current_Transformers
+15V
-15V
GND
Repeat(+15V)
Repeat(-15V)
Repeat(GND)
Chose size of UZ logo by chosing footprint type
UltraZohm
LOGO1
INFO1
ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv
ZC Serialnumber
Place MH close to edge connector when ordering PCB (without assembly) from JLC. They ensure accurate outline milling of the edge connector.
PS_Signal_Conditioning
Power_Supply_Signal_Conditioning
INFO2
Project
ProjectRevision
AuthorParam
ProjectDate
Design Information
Mounting Holes
MH1 3.4mm Mounting Hole
MH2 3.4mm Mounting Hole
MH3 3.4mm Mounting Hole
MH4 3.4mm Mounting Hole
Title TopSheet.SchDoc
Revision: 01
Design Engineer: Tom Gaupp
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb
UltraZohm
www.ultrazohm.com
Date: 27/05/2026
Sheet 1 of 12



Signal Conditioning 1..4

Verstärkerschaltung Gain: G_amp = 2

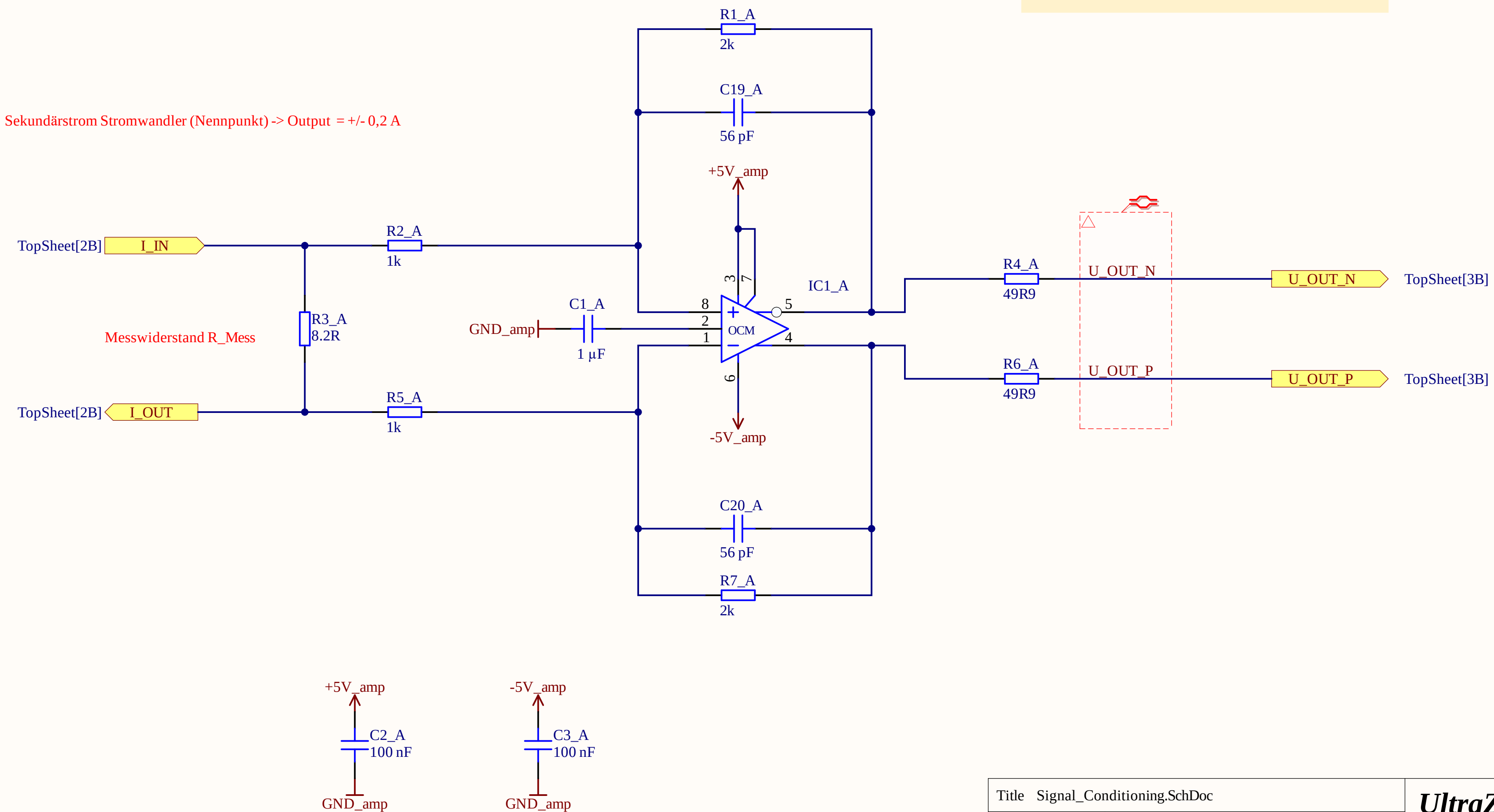
G_amp = R_F / R_G

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: fg

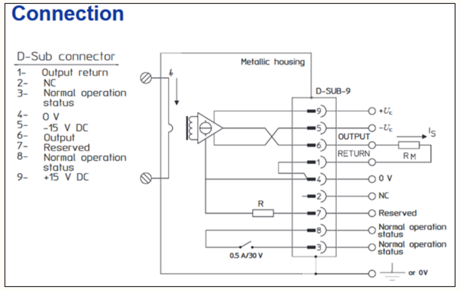
Rückkopplungskondensator: Cf =

Cf = 1 / 2*pi*Rf*Cf = ... =



Title Signal_Conditioning.SchDoc	*UltraZohm* www.ultrazohm.com
Revision: 01 Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb	Date: 27/05/2026 Sheet 2.1 of 12

Signal Conditioning 1..4



Verstärkerschaltung Gain: G_amp = 2

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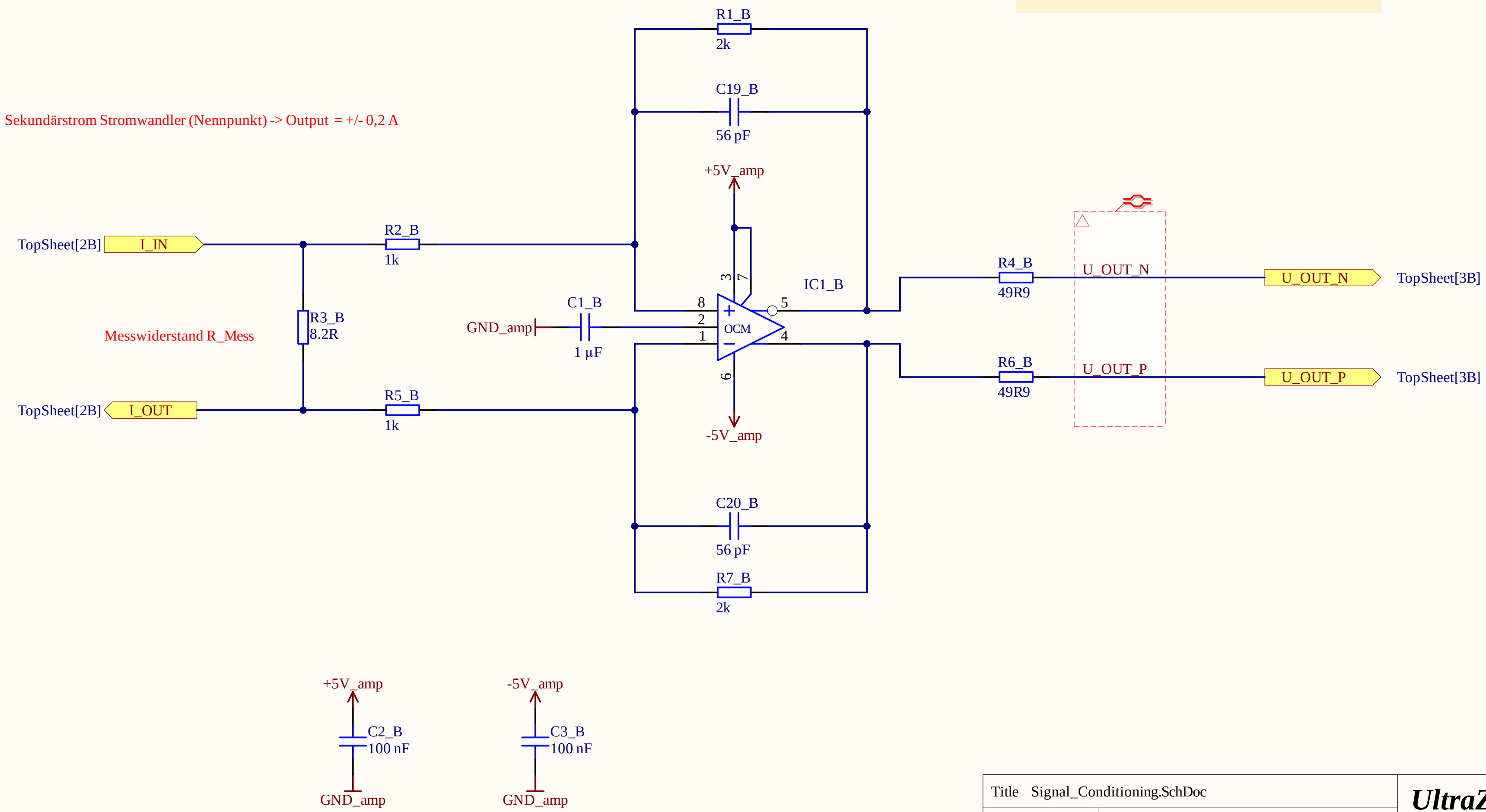
Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: fg

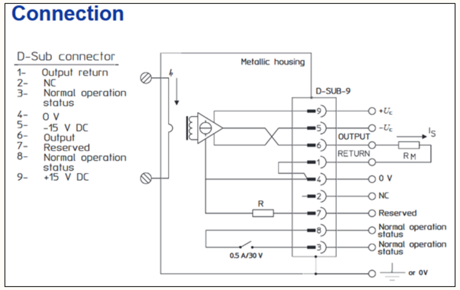
Rückkopplungskondensator: Cf =

Cf = 1 / 2*pi*Rf*Cf = ... =



Title Signal_Conditioning.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: 01 Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb	Date: 27/05/2026 Sheet 2.2 of 12

Signal Conditioning 1..4



Verstärkerschaltung Gain: G_amp = 2

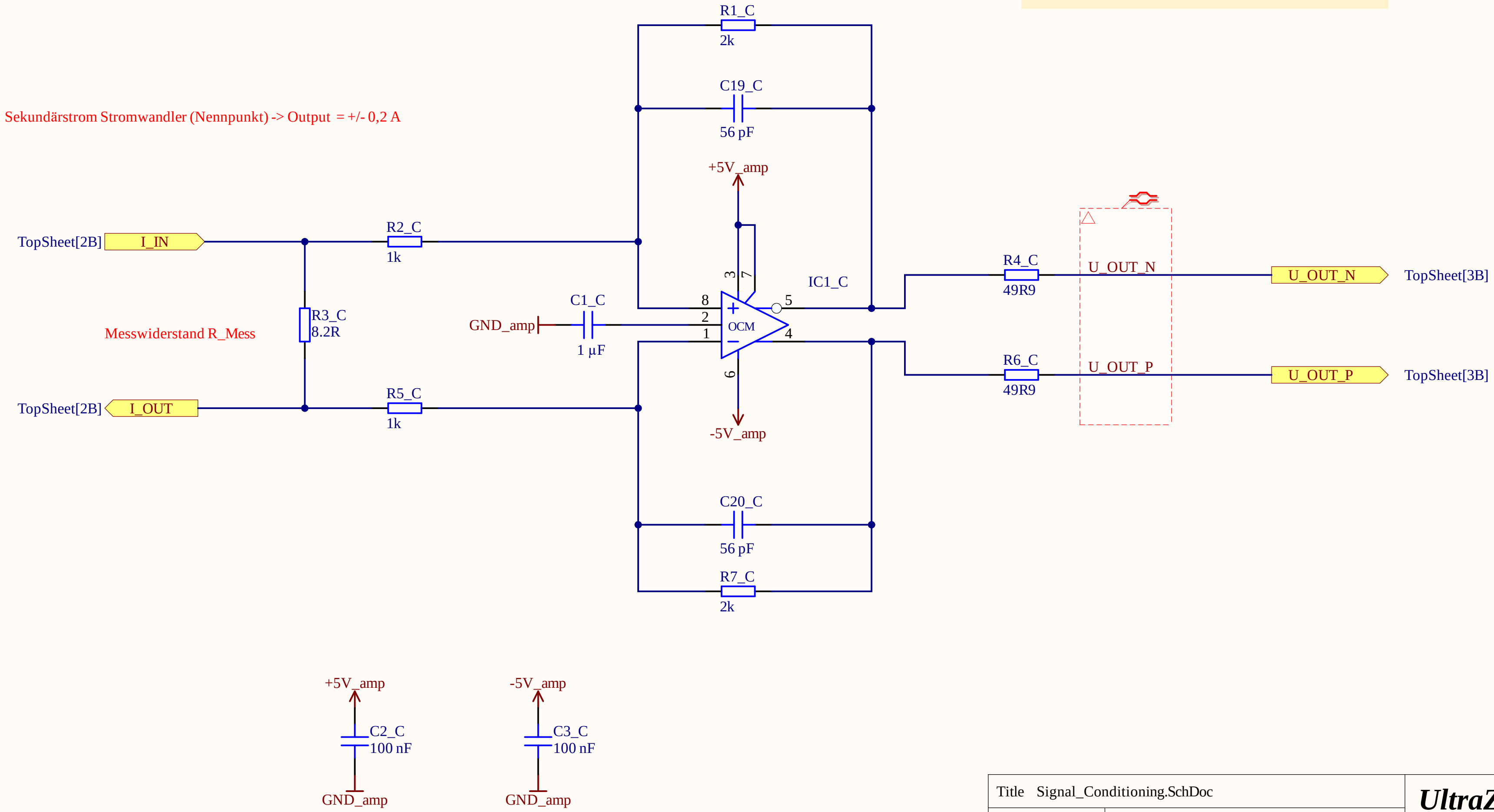
G_amp = R_F / R_G

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

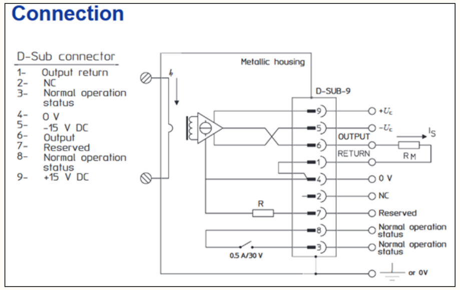
Grenzfrequenz: fg

Rückkopplungskondensator: Cf =

Cf = 1 / 2*pi*Rf*Cf = ... =



Title Signal_Conditioning.SchDoc		***UltraZohm*** www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 2.3 of 12



Signal Conditioning 1..4

Verstärkerschaltung Gain: G_amp = 2

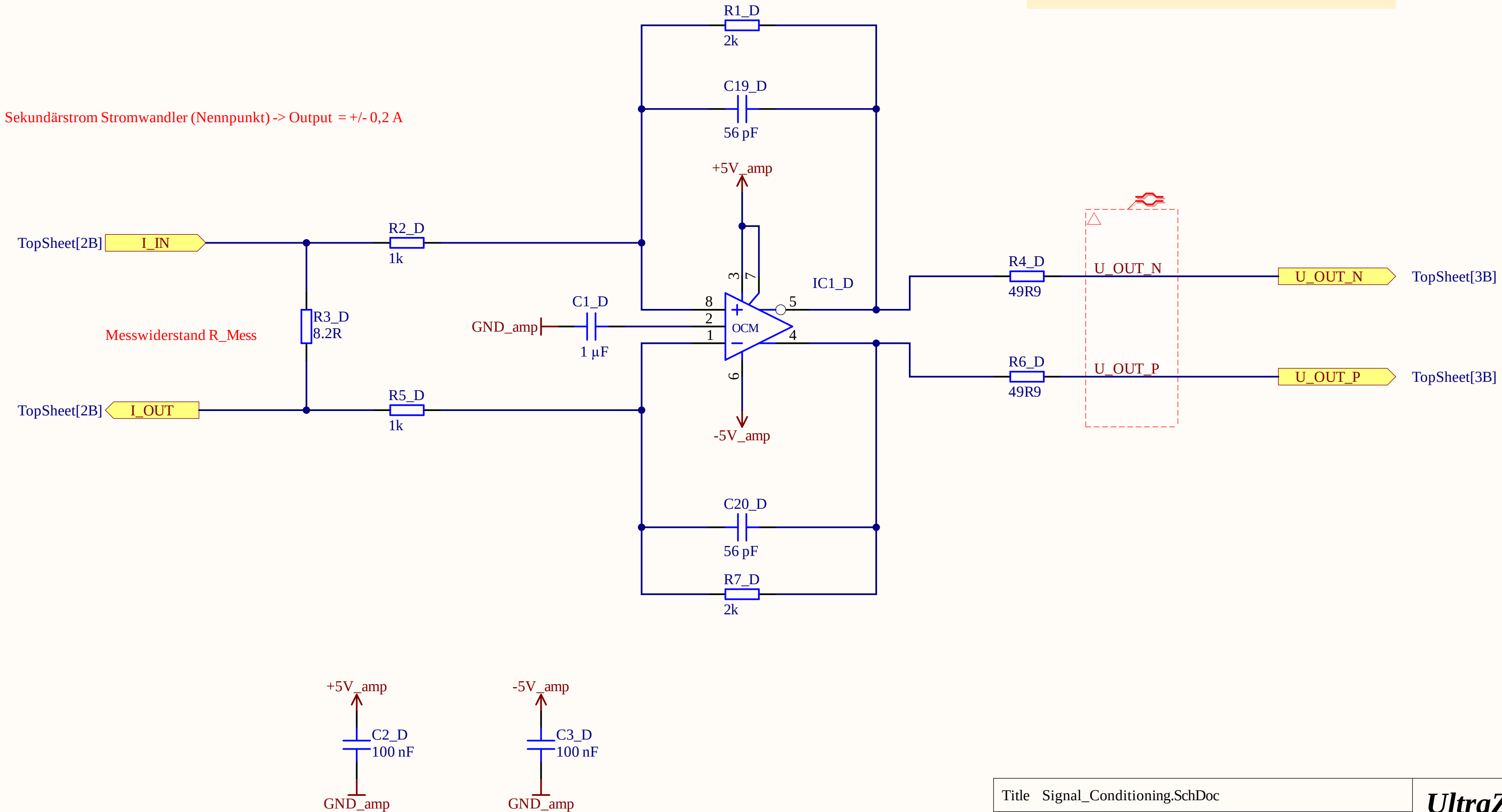
G_amp = R_F / R_G

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: fg

Rückkopplungskondensator: Cf =

Cf = 1 / 2*pi*Rf*Cf = ... =



Title	Signal_Conditioning.SchDoc
Revision: 01	Design Engineer: Tom Gaupp
Project:	uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm
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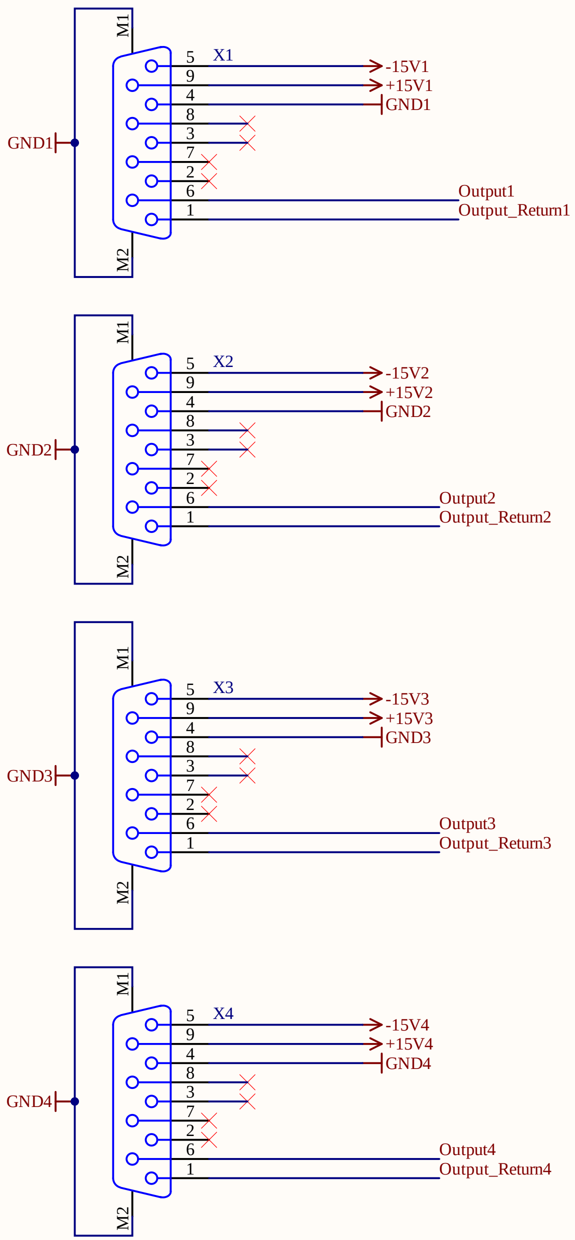
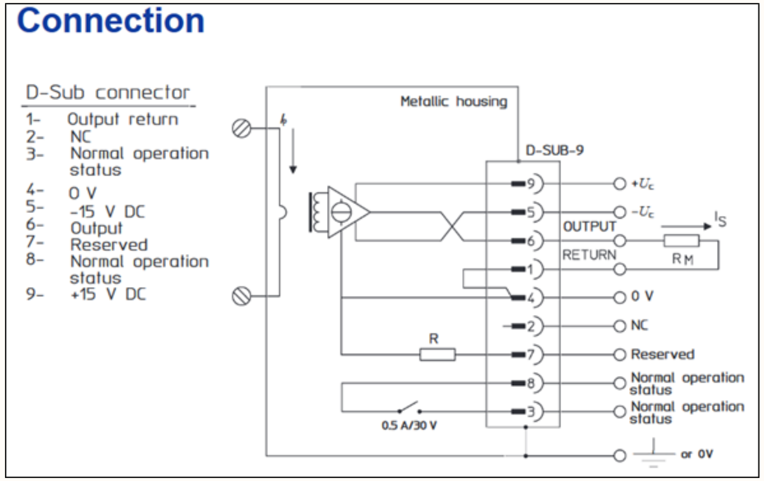
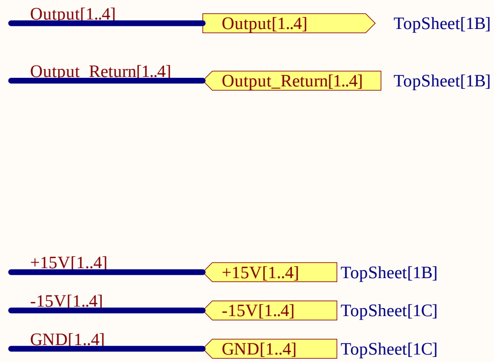
Date: 27/05/2026
Sheet 2.4 of 12

Analog Input







Title DSub_Analog_Input.SchDoc		**UltraZohm** www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 3 of 12

Analog Output

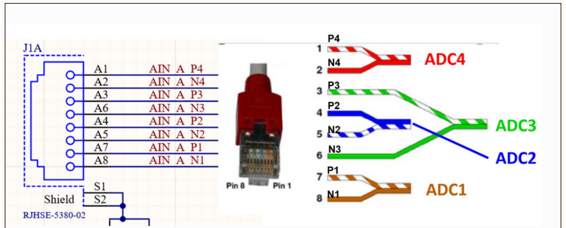
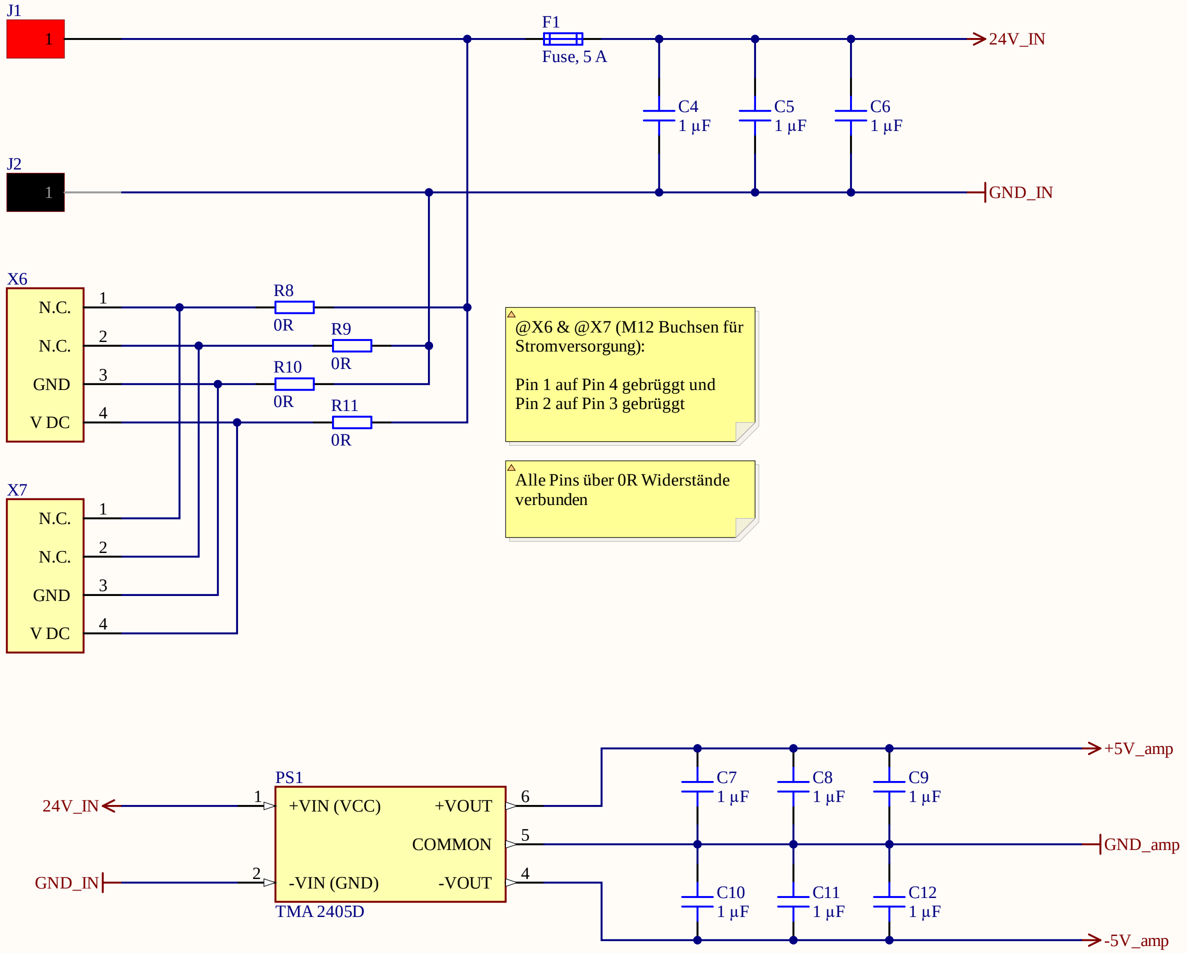


Fig. 186 *Pinout of the ADC ethernet port.*

Title RJ45_Analog_Output.SchDoc		**UltraZohm** www.ultrazohm.com	
Revision: 01	Design Engineer: Tom Gaupp		
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 4 of 12	

Power Supply for Amplification Circuit



Quelle: DC/DC Wandler 24V auf +/- 5V

Verbraucher: ADA4940 fully differential Amplifier

Ruhestrom:	**1,25 mA**
bei	**5,0 V**
Power:	**0,00625 mW**

Output Characteristics:

U_out ~ Vs = 5 V
I_out = 46 mA peak
P _out max = 0,23 W

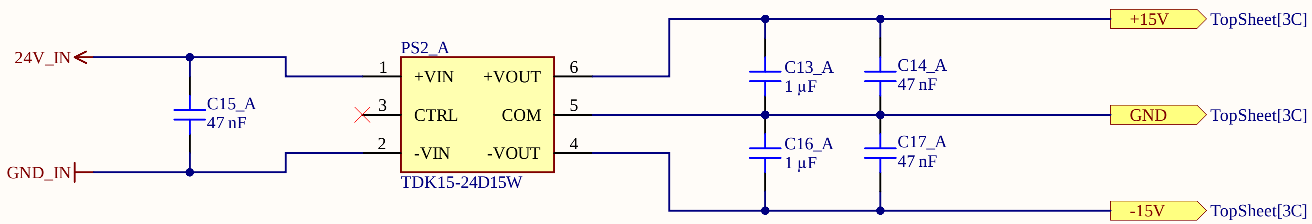
Title Power_Supply_Signal_Conditioning.SchDoc		UltraZohm www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 5 of 12

Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
DC Nennstromaufnahmen @15V = 320 mA
I_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A
Shielded output cable & shielded plug are recommended to use
____-
Max Power Consumption (pro Stromsensor):
P = U*I_max = 15 V * 0,420 A = 6,3 W



Title Power_Supply_Current_Transformers.SchDoc		*UltraZohm* www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 6.1 of 12

Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
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I_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A
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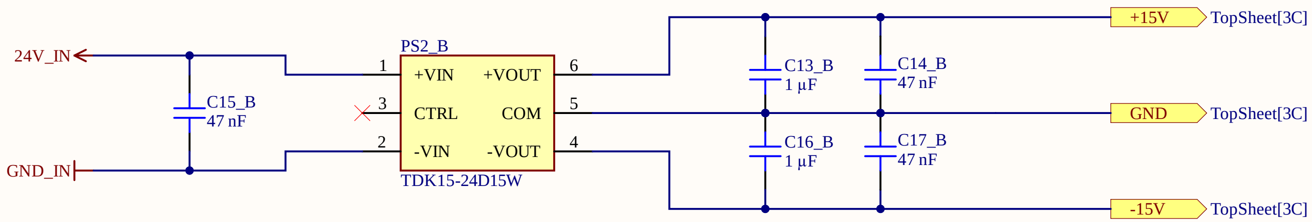
P = U*I_max = 15 V * 0,420 A = 6,3 W



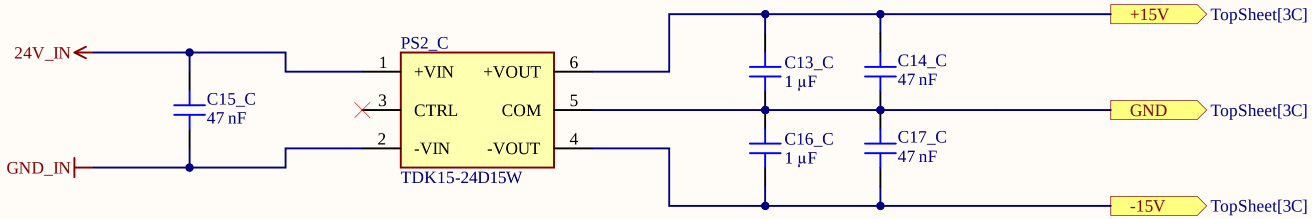
Title Power_Supply_Current_Transformers.SchDoc		***UltraZohm*** www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 6.2 of 12

Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
DC Nennstromaufnahmen @15V = 320 mA
I_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A
Shielded output cable & shielded plug are recommended to use
____-
Max Power Consumption (pro Stromsensor):
P = U*I_max = 15 V * 0,420 A = 6,3 W



Title Power_Supply_Current_Transformers.SchDoc		**UltraZohm** www.ultrazohm.com
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 6.3 of 12

Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
DC Nennstromaufnahmen @15V = 320 mA
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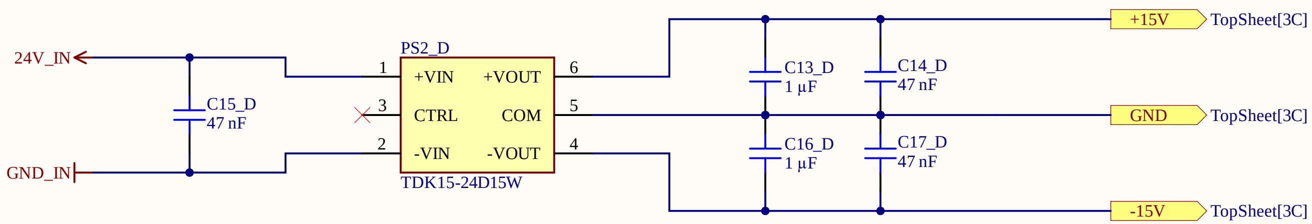
____-
Max Power Consumption (pro Stromsensor):
P = U*I_max = 15 V * 0,420 A = 6,3 W



Title Power_Supply_Current_Transformers.SchDoc		***UltraZohm***
Revision: 01	Design Engineer: Tom Gaupp	www.ultrazohm.com
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb		Date: 27/05/2026 Sheet 6.4 of 12